REQUIREMENT ANALYSIS DOCUMENT

Problem Statement: Education-Oriented Financial & Paper Trading Platform

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Course: B.TECH - Computer Science & Engineering
Institution: IIIT SRICITY
Project: FFSD PROJECT
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Academic Year: 2026-2027

Introduction

Purpose:

The purpose of this document is to define the functional and non-functional requirements of the *Education-Oriented Financial Literacy and Paper Trading Platform*. This document serves as a reference for developers, instructors, and evaluators to understand system behavior, constraints, and expected features.

Scope of the project:

The system is a web-based platform that enables students to learn basic financial concepts and practice stock trading using virtual money. It includes learning modules, paper trading simulation, portfolio tracking, instructor dashboards, and administrative tools.

Definitions:

Term	Meaning
Paper Trading	Simulated trading using virtual money
Portfolio	Collection of virtual assets held by a user
Admin	System administrator
Instructor	Faculty user monitoring students
Trading Assignment	A practical trading task assigned to learners for skill development
Enrollment	The association of a learner with a course
Market Price Feed	Source of financial instrument price information used in trading simulation
Course Module	A learning unit containing educational content, exercises, and assessments
Financial Instrument	A tradable market asset such as stocks, ETFs, or indices available in simulation

Term	Meaning
Matching Engine	Core component responsible for matching buy and sell orders based on price and time priority
Order Book	Data structure holding all pending buy and sell orders for a stock
Market Order	Order that executes immediately at the best available price
Limit Order	Order that executes only at a specified price or better
Stop-Loss Order	Order that executes when the market price reaches a predefined value
Partial Fill	When only part of an order quantity is matched
Cash Check	Verification that a user has enough balance before placing an order
Margin Check	Verification that margin requirements are satisfied (if margin trading is enabled)
Order Book Snapshot	Periodic saved state of the order book for recovery
Asynchronous Database Writes	Writing trade and portfolio updates to the database in the background
Latency	Delay in market data updates (considered non-critical for learning systems)

DESCRIPTION

Product Perspective:

The platform is a standalone web application developed using a full-stack architecture. It integrates a frontend user interface, a backend server for business logic, and a database for persistent storage. Market price data may be fetched from a third-party API or a static dataset.

USER CLASSES:

User Type	Description
Learner	Learn Financial concepts and performs paper trading
Instructor	Assigns tasks, monitors student performance
Course provider	Provide courses and create assignments
Admin	Manages users, learning content, and stock data

Operating Environment:

The system will work on common web browsers like Chrome, Firefox, and Edge, so users don’t need to install anything special.The main program will run on a computer or an online server that handles all the background work.All the user data and app information will be stored safely in a database system.

User Requirements

Learner Requirements:

- Learners should be able to view real-time or near real-time stock prices.
- Learners should be able to buy and sell stocks using virtual money.
- Learners should be able to view their virtual portfolio and transaction history.
- Learners should be able to track profit/loss and overall performance, including simulated brokerage charges and capital gains taxes.
- Learners should be able to access basic financial learning modules and guidelines.

Instructor Requirements:

- Instructors should be able to view learner's performance and portfolios.
- Instructors should be able to assign tasks or trading challenges.
- Instructors should be able to monitor course progress and task completion status.
- Instructors should be able to provide feedback or remarks.

Admin Requirements:

- Admin should be able to manage users (add, delete, block, update).
- Admin should be able to manage stock data sources and APIs.
- Admin should be able to manage learning content.
- The admin shall be able to configure trading rules such as virtual capital limits, market hours, allowed instruments, and risk constraints.

- The Admin shall be able to enable, disable, or restrict platform features based on academic or operational requirements.

Course Provider Requirements:

- The Course Provider shall be able to create and submit course offerings including course title, description, learning objectives, duration, and pre-requisites.

- The Course Provider shall be able to update or revise course content based on market changes, learner feedback, or academic requirements.

- The Course Provider shall be able to publish updates or announcements related to course content or schedule changes.

- The Course Provider shall be able to manage assessments and evaluation components associated with each course.

- The Course Provider shall be able to define scoring and ranking mechanisms for trading challenges.

- The Course Provider shall be able to integrate challenge results into course assessments.

Functional Requirements

Trading Simulation:

- The system shall display stock prices.
- The system shall allow buying and selling stocks using virtual money.
- The system shall update the learner’s portfolio in real time.

Portfolio Management:

- The system shall display portfolio value.
- The system shall calculate profit/loss.
- The system shall store transaction history.

Learning Module:

- The system shall display educational content.
- The system shall allow course provider to add learning content.
- The system shall track content completion.

Admin Management:

- The system shall allow admin to manage users.
- The system shall log system activities.

Non-Functional Requirements

Performance:

- The system should respond within 2 seconds for user actions.
- The system should handle at least 500 concurrent users(for now).

Security:

- The system should encrypt user passwords.
- The system should use HTTPS.
- The system should prevent unauthorized access.

Usability:

- The system should be easy to use for beginners.
- The system should support both desktop and mobile browsers.

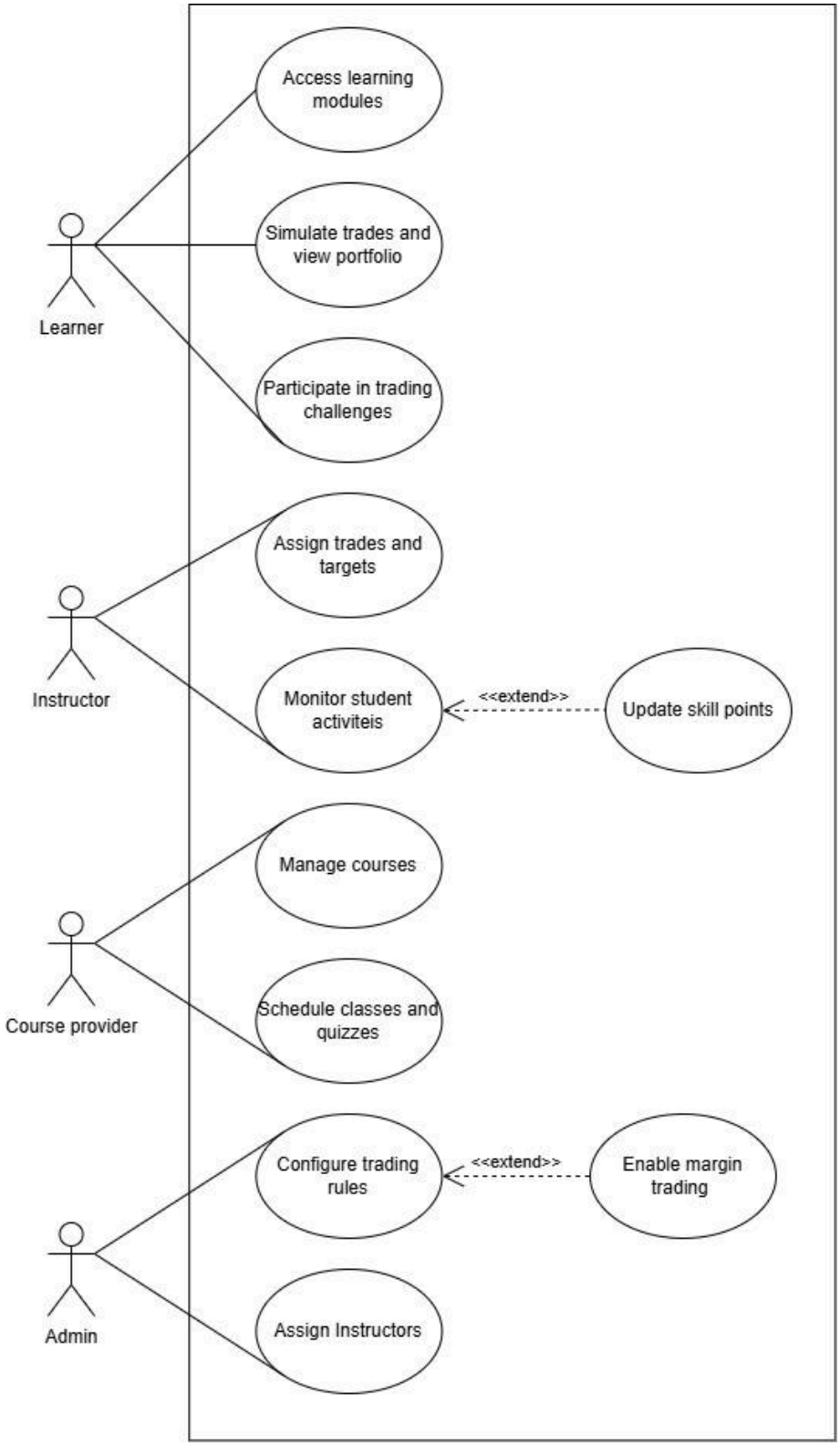
Reliability:

- The system should have at least 99% uptime.
- The system should handle failures gracefully.

Scalability:

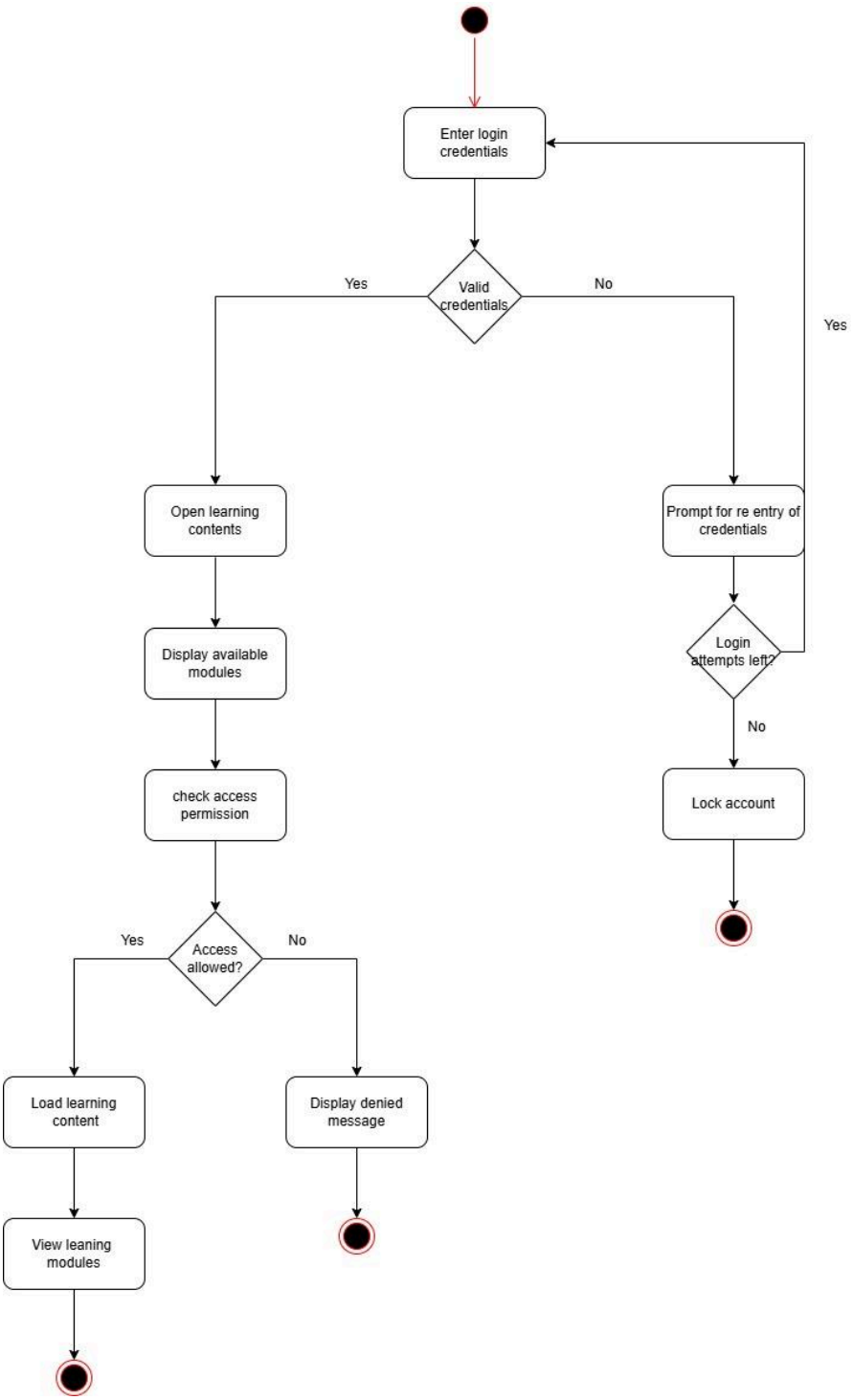
- The system should support future feature expansion.
- The system should handle increased user load.

USE CASE DIAGRAM:

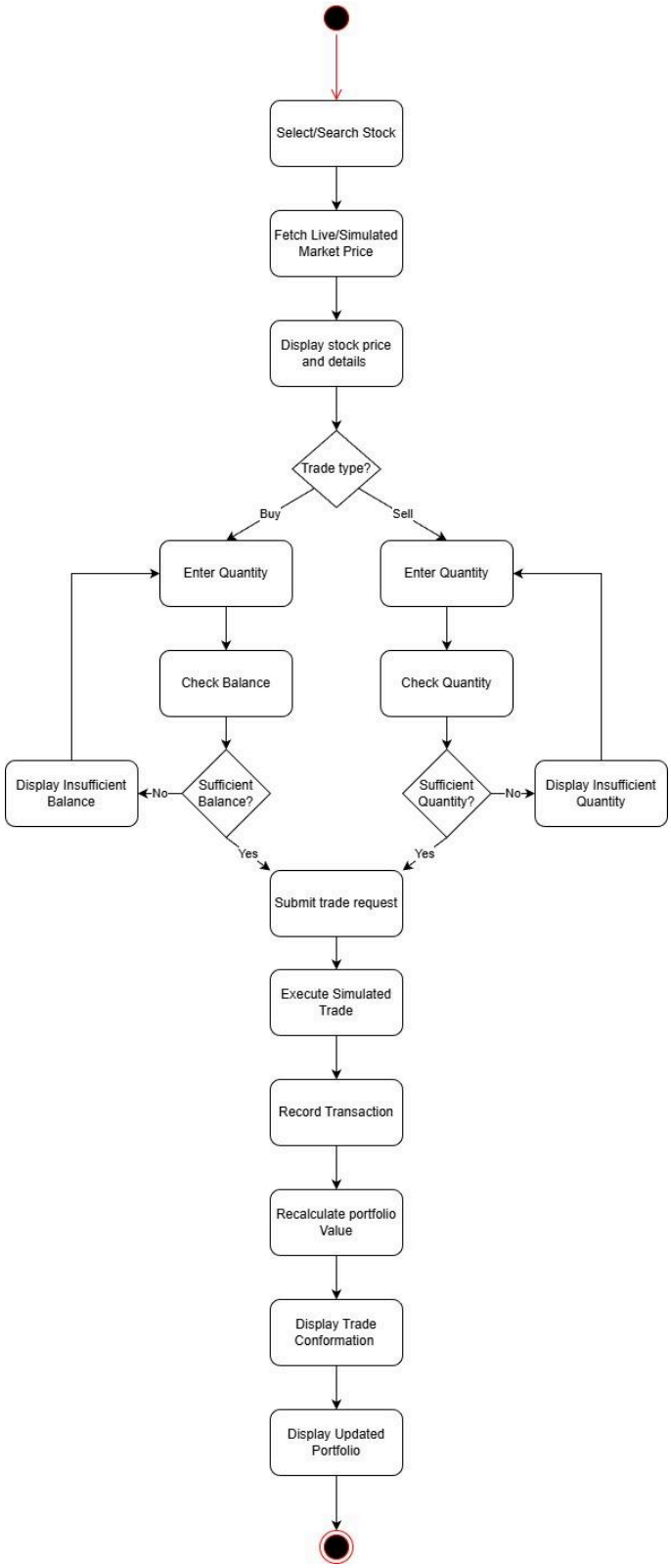


Activity Diagrams

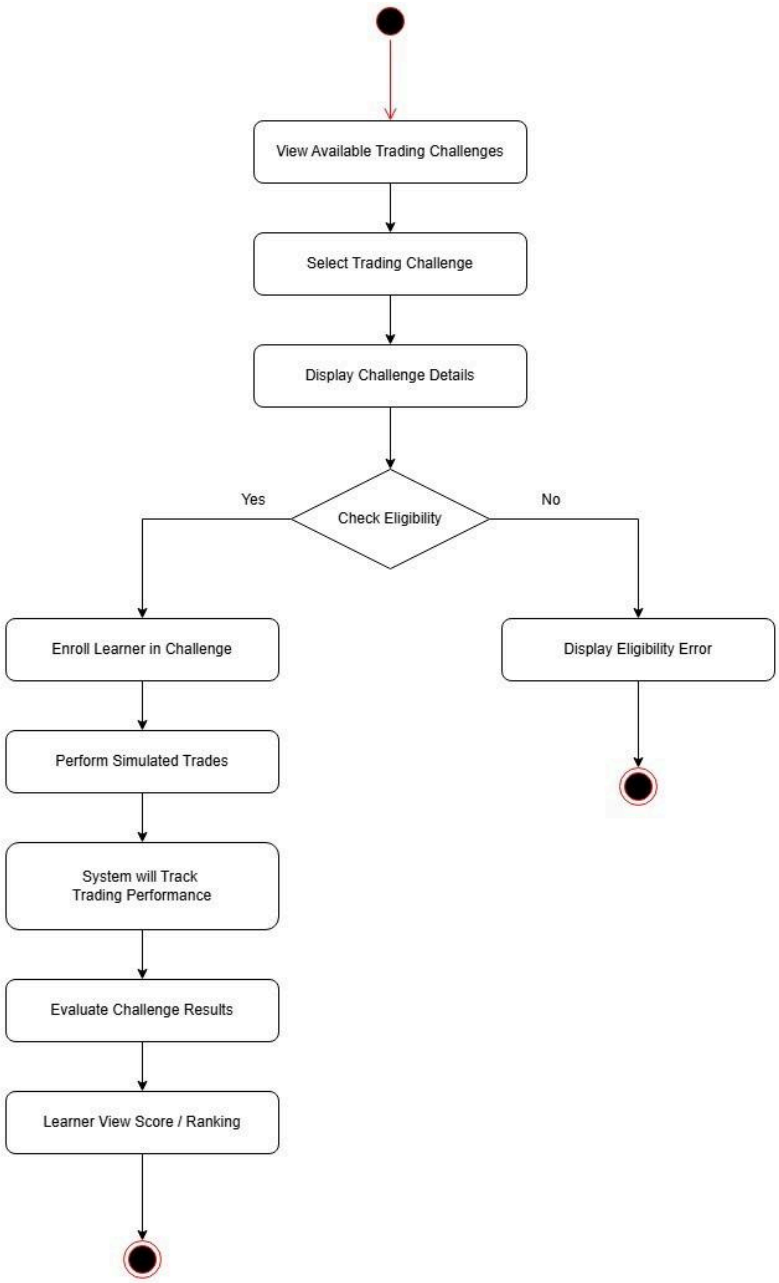
1. Access Learning modules



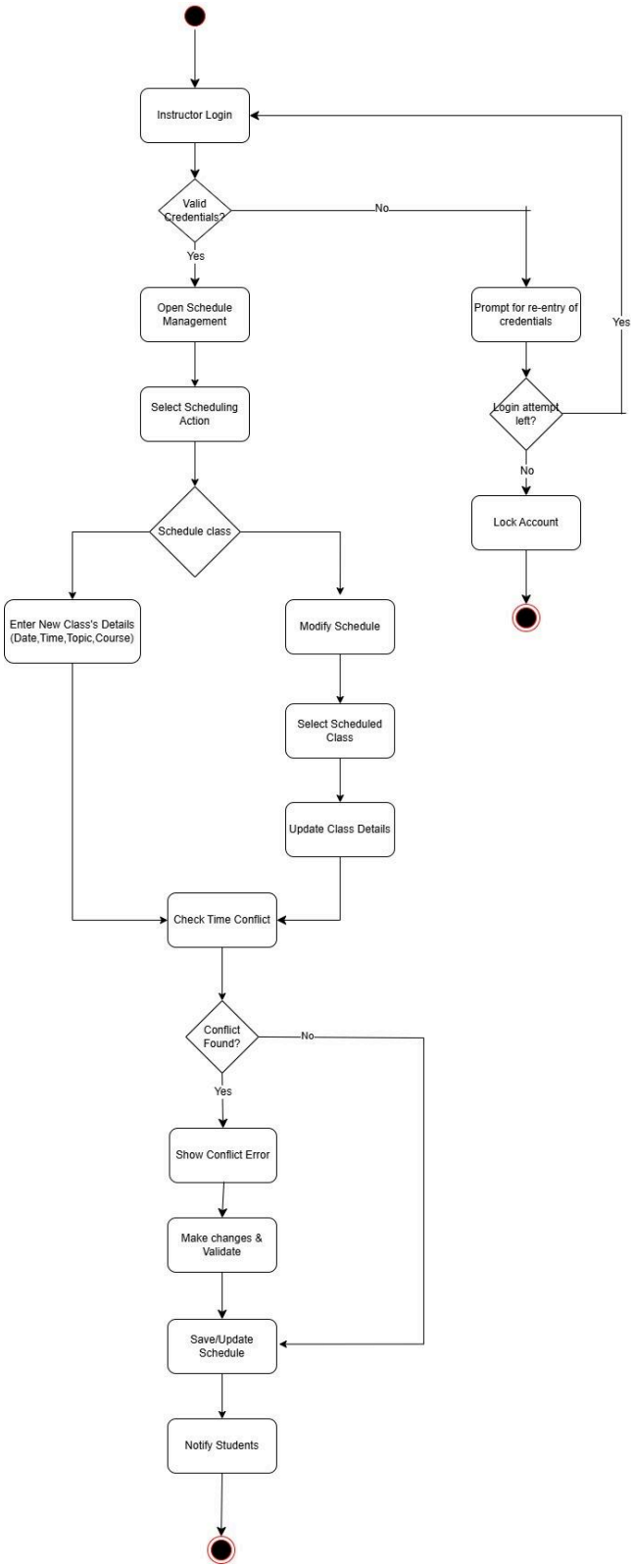
2. Simulate trades



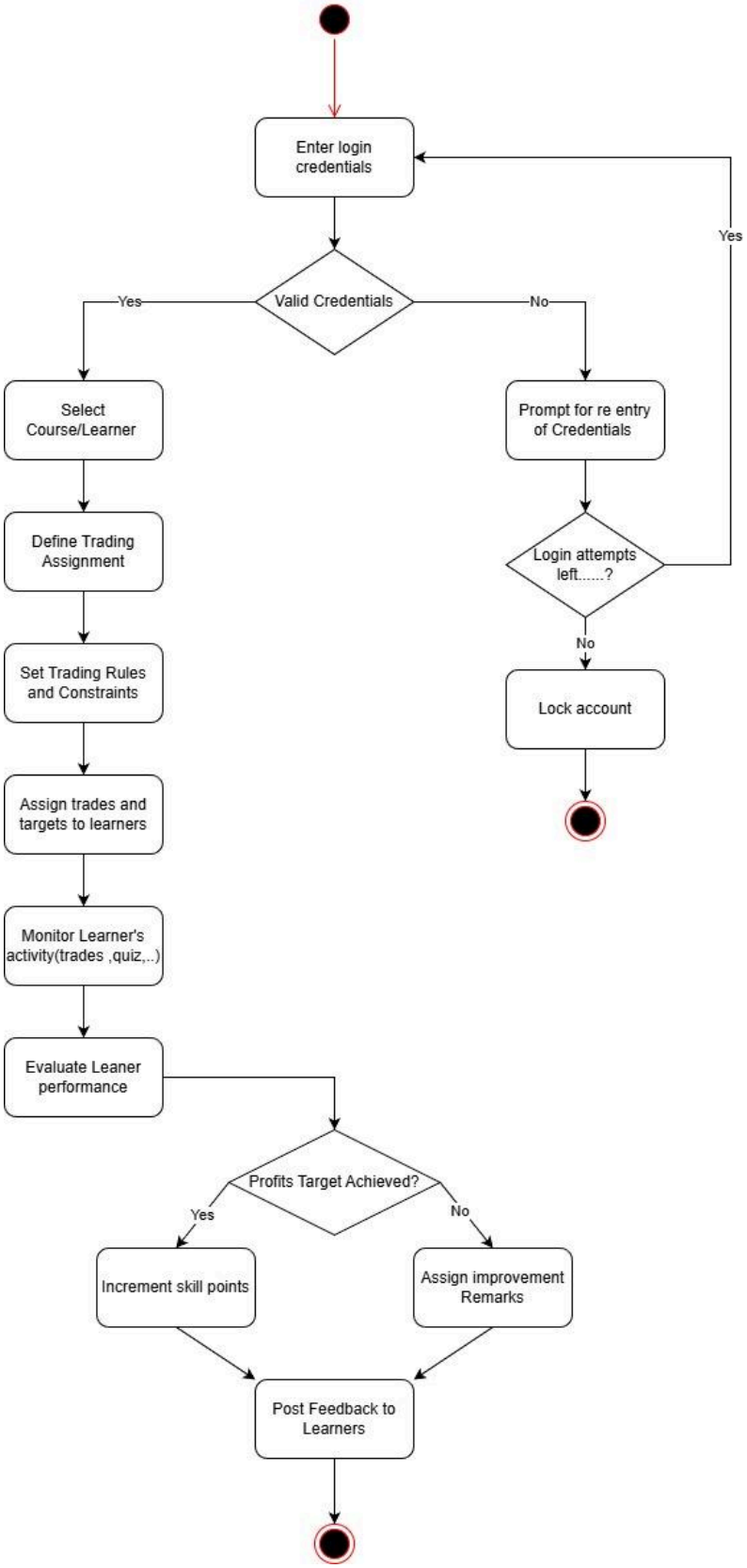
3.Participate in trading challenges



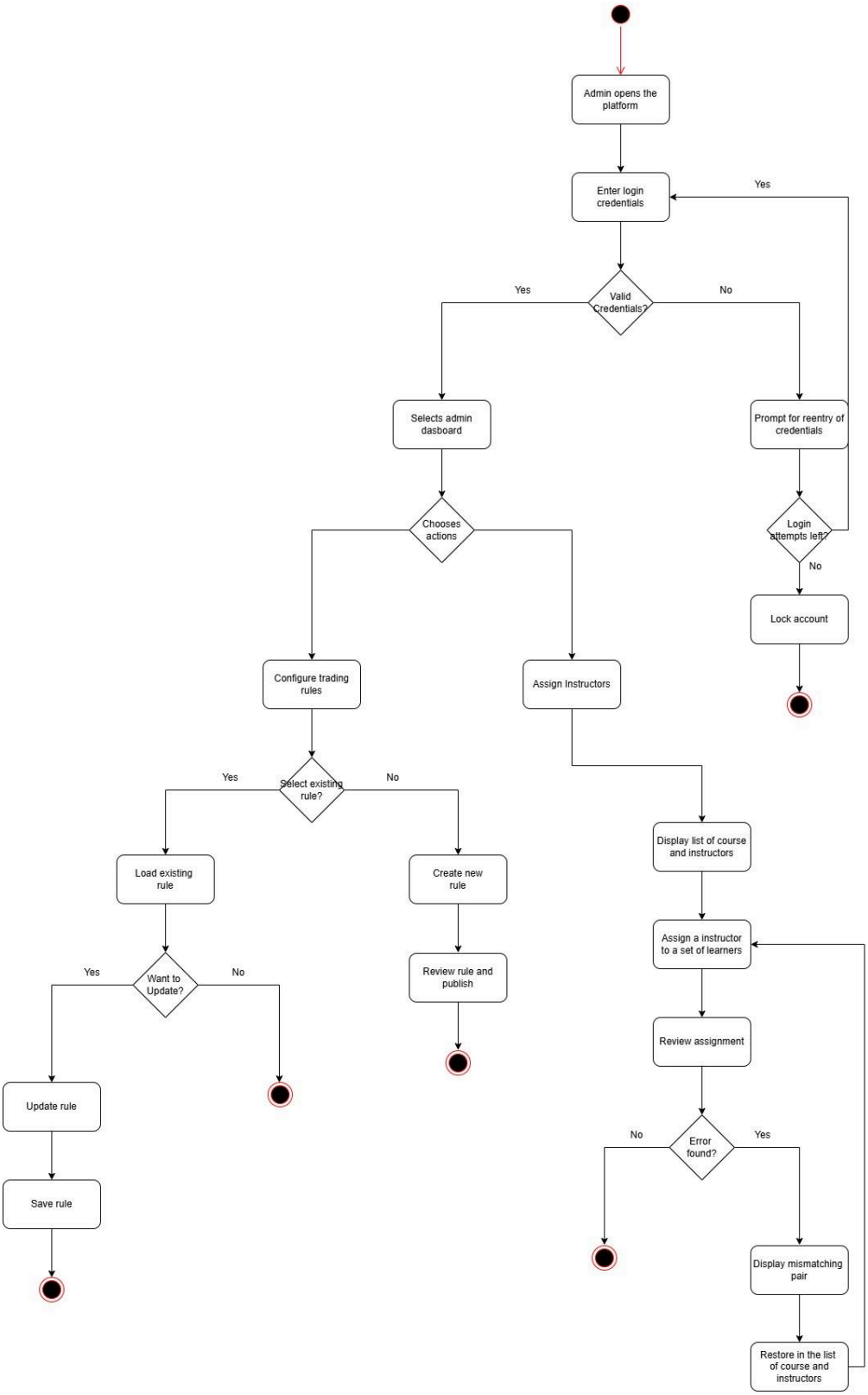
4.Schedule Classes



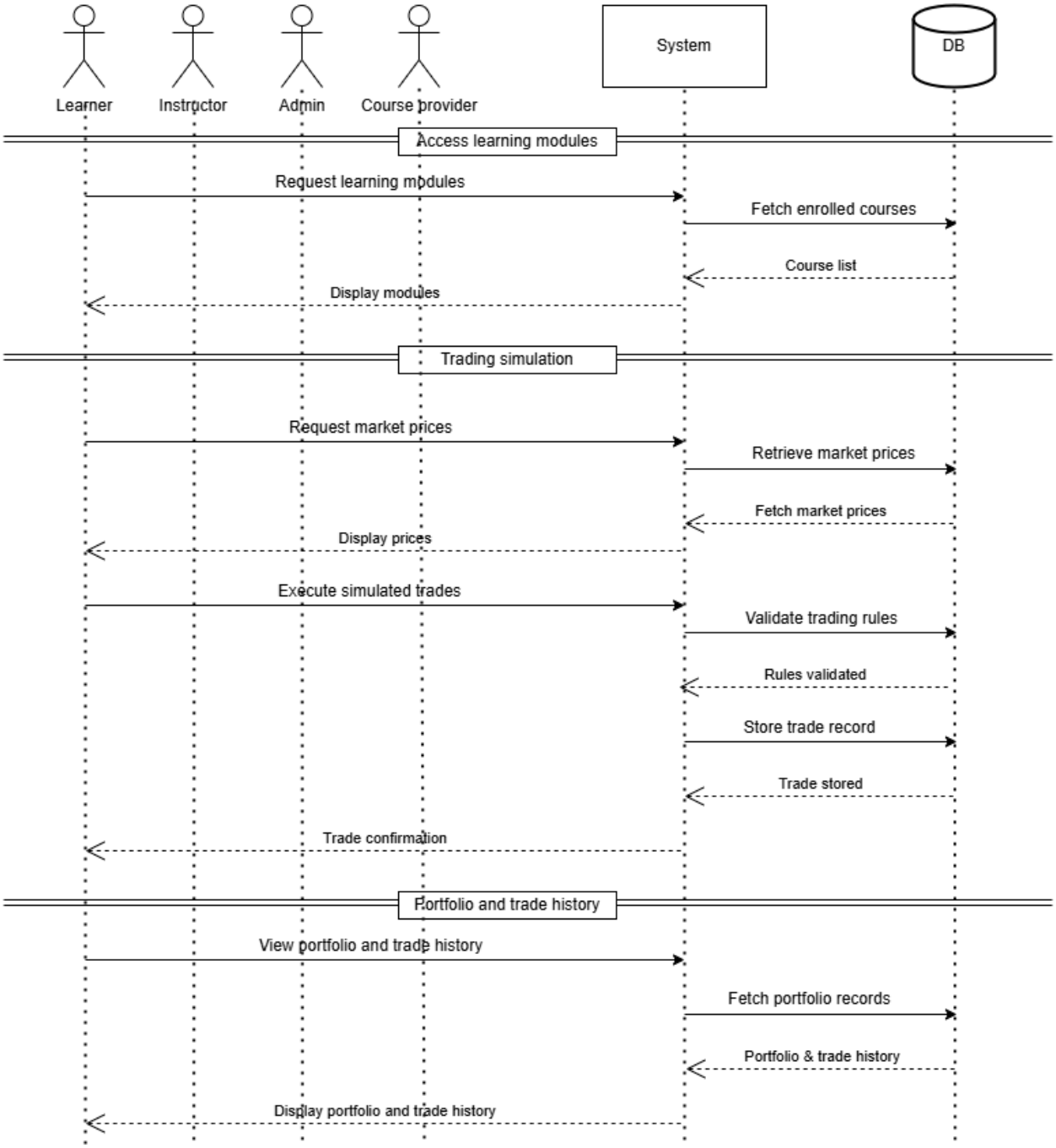
5.Instructor workflow

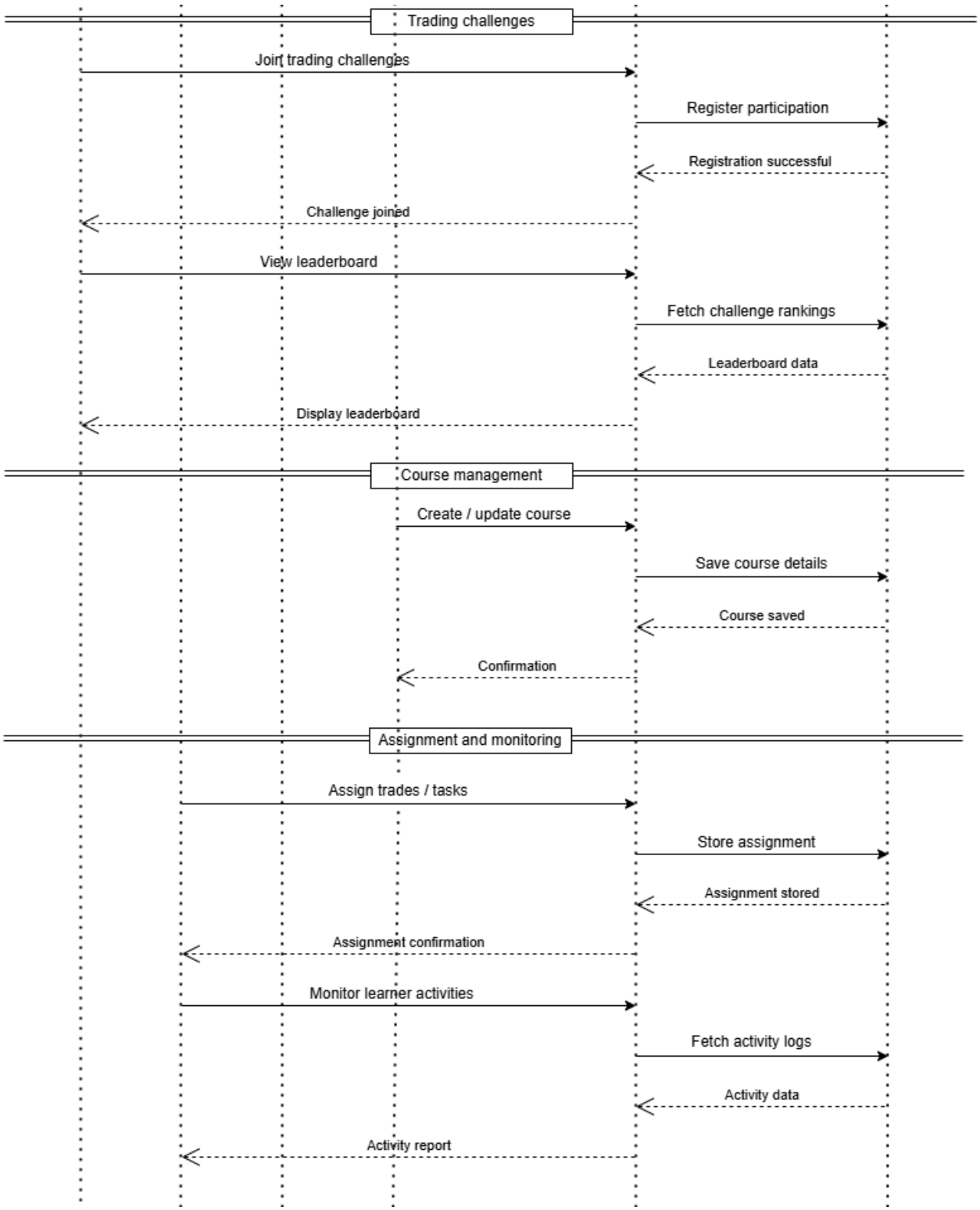


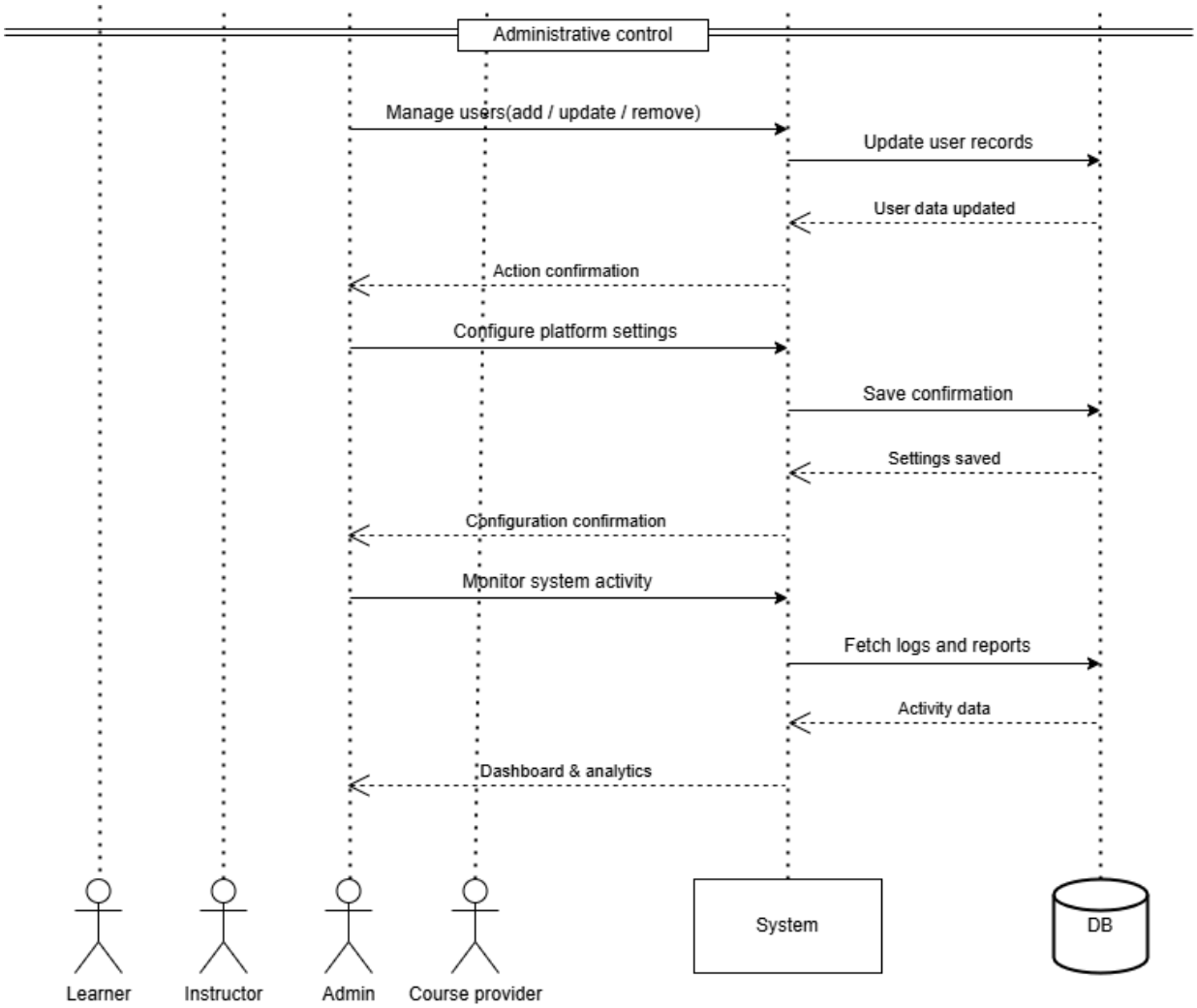
6.Admin workflow



Sequence Diagram




Trading challenges
Join trading challenges
Register participation
Registration successful
Challenge joined
View leaderboard
Fetch challenge rankings
Leaderboard data
Display leaderboard
Course management
Create / update course
Save course details
Course saved
Confirmation
Assignment and monitoring
Assign trades / tasks
Store assignment
Assignment stored
Assignment confirmation
Monitor learner activities
Fetch activity logs
Activity data
Activity report


Administrative control
Manage users(add / update / remove)
Update user records
User data updated
Action confirmation
Configure platform settings
Save confirmation
Settings saved
Configuration confirmation
Monitor system activity
Fetch logs and reports
Activity data
Dashboard & analytics
Learner
Instructor
Admin
Course provider
System
DB